

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T45 · Volume and Surface Area

With Answers

Questions with full worked solutions

■ 40 pages

FAMILY MAP

Volume Surface Area

F01	Find volume of prisms and cylinders	7 Q 30 marks
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F02	Find surface area of prisms and cylinders	4 Q 15 marks
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F03	Resolve composite solids and volume-unit constraints	5 Q 23 marks
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F04	Use sphere and cone formulae	2 Q 9 marks
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- 7 Here is a solid cylinder.

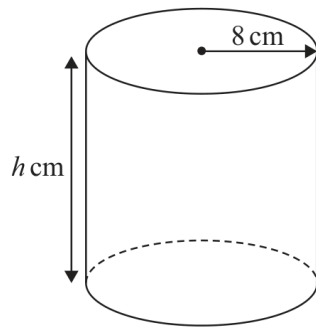


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 7 is 3 marks)

Worked Solution 05

Volume of a cylinder

UNIT 2**3 marks****Use the volume formula**

$$3892 = \pi(8)^2 h = 64\pi h.$$

Solve and round

$$h = \frac{3892}{64\pi} = 19.357219 \dots \Rightarrow h = 19.4 \text{ cm (1 d.p.)}.$$

Final answer

$$h = 19.4 \text{ cm}$$

21 The diagram shows a cuboid.

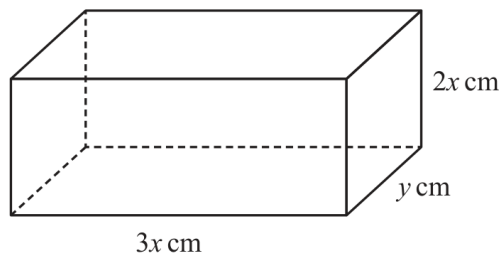


Diagram **NOT**
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A\text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

Worked Solution 30

Surface area of a cuboid

UNIT 2
3 marks**Use volume**

$$6x^2y = 1014 \Rightarrow y = \frac{169}{x^2}.$$

Write surface area

$$A = 2(6x^2 + 3xy + 2xy) = 12x^2 + 10xy.$$

Substitute

$$A = 12x^2 + 10x \left(\frac{169}{x^2} \right) = 12x^2 + \frac{1690}{x}.$$

Final answer

$$A = 12x^2 + \frac{1690}{x}$$

Question 10

4MA1-1H Q23

ORIGINAL QUESTION

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

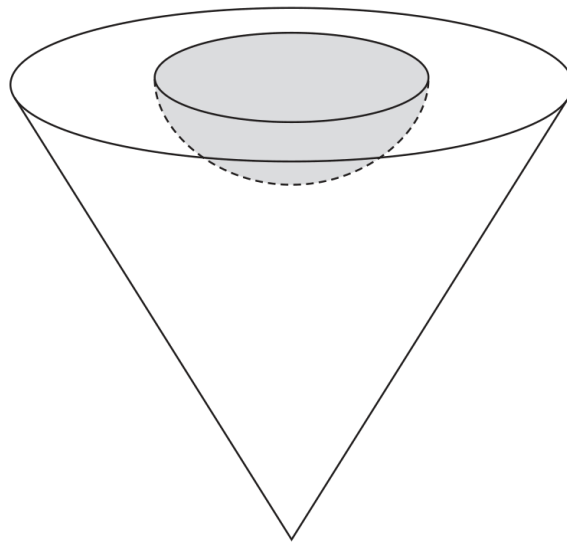


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

Worked Solution 10

Cone with a hemisphere removed

MODULAR UNIT 2

5 marks

Find the scale

$$50\pi x^3 - \frac{16}{3}\pi x^3 = 6948\pi \Rightarrow x^3 = 10422/67.$$

Removed surface

$$3\pi(2x)^2 = 12\pi x^2 = 12\pi(10422/67)^{2/3} = 1090.390\dots$$

Final answer

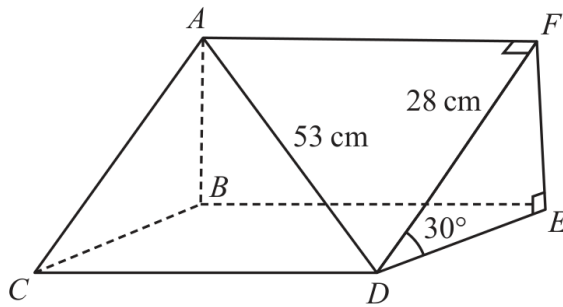
$$1090 \text{ cm}^2$$

Question 16

4MA1-1H Q21 demand

ORIGINAL QUESTION UNIT 2

5 marks

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

Worked solution 16

Volume of a triangular prism

MODULAR UNIT 2

5 marks

Demand – Volume of a triangular prism

Resolve the triangular end

$$FE = 28 \sin 30^\circ = 14, \quad DE = 28 \cos 30^\circ = 14\sqrt{3}.$$

Find prism length

$$AF = \sqrt{53^2 - 28^2} = 45.$$

Calculate volume

$$V = \frac{1}{2}(14)(14\sqrt{3})(45) = 4410\sqrt{3} \approx 7638.34.$$

Final answer7638 cm³

Original question 04

4MA1-1H Q5 M01

MODULAR UNIT 2

4 marks

- 5 The diagram shows a cylinder.

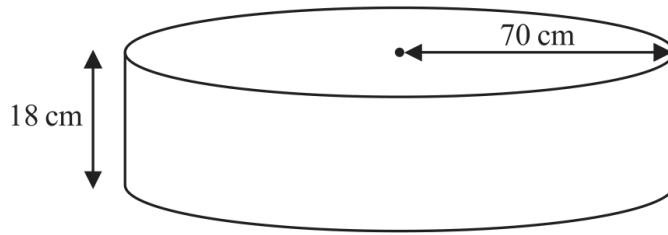


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

..... litres

Worked solution 04

Volume of a cylinder in litres

MODULAR UNIT 2

4 marks

M01 – Volume of a cylinder in litres**Use the cylinder formula**

$$V = \pi r^2 h = \pi (70)^2 (18) = 88200\pi \text{ cm}^3.$$

Convert cm^3 to litres

$$V = 88200\pi / 1000 = 277.088 \dots \text{ litres.}$$

Round as requested

$$V \approx 277 \text{ litres.}$$

Final answer

277 litres

Original question 14

4MA1-1H Q22 M01

MODULAR UNIT 2

3 marks

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

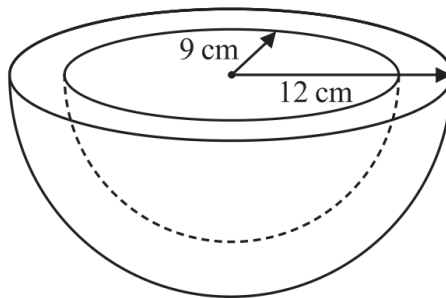


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

Worked solution 14

Find the volume of metal in a hemispherical bowl

MODULAR UNIT 2

3 marks

M01 – Find the volume of metal in a hemispherical bowl**Subtract hemisphere volumes**

$$V = \frac{2}{3}\pi(12^3 - 9^3).$$

Evaluate

$$V = 2091.5 \dots \text{ cm}^3.$$

Round

$$V = 2092 \text{ cm}^3.$$

Final answer

$$2092 \text{ cm}^3$$

Original question 20

4MA1-2H Q6 M01

MODULAR UNIT 2

3 marks

- 6 The diagram shows a solid triangular prism.

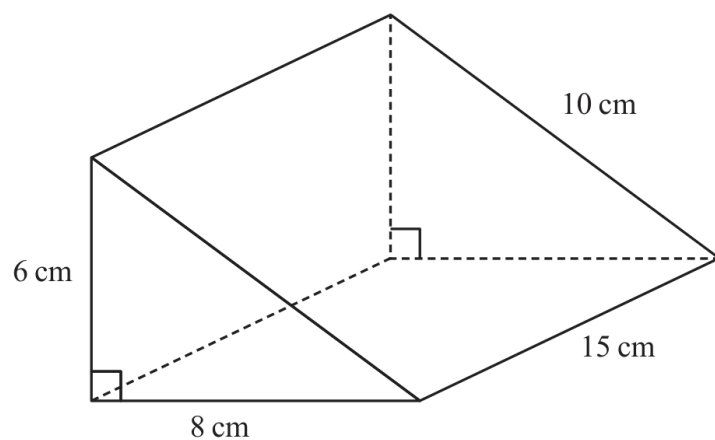


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

Worked solution 20

Surface area of a triangular prism

MODULAR UNIT 2

3 marks

M01 – Surface area of a triangular prism**Find the triangular ends**

$$2 \left(\frac{1}{2} \times 8 \times 6 \right) = 48.$$

Find the rectangular faces

$$15(8) + 15(6) + 15(10) = 360.$$

Add

$$48 + 360 = 408 \text{ cm}^2.$$

Final answer

$$408 \text{ cm}^2$$

Original question 08

4MA1-1H Q7 Q7

ORIGINAL QUESTION**4 marks**

- 7 A cylinder is placed on the ground.

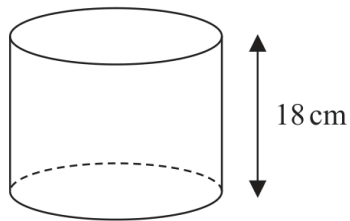


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

Worked solution

Family: Find volume of prisms and cylinders • Variant: Find volume of a cuboid or cylinder

METHOD / WORKING

MODULAR UNIT 2 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 08 MODULAR UNIT 2
Chapter: Volume & Surface Area 4 marks
Q7 – Chapter: Volume & Surface Area
Family: Find volume of prisms and cylinders • Variant: Find volume of a cuboid or cylinder
Method
Using pressure = force/area, $1.4 = 72/A$.
Why: Substitute the given pressure and force into the stated formula.
Method
<i>The circular base area is</i> $A = 72/1.4 = 51.428571 \dots \text{cm}^2$.
Why: Rearrange the pressure formula for area.
Method
<i>The cylinder volume is</i> $\text{base area} \times \text{height} = 51.428571 \dots \times 18 = 925.714 \dots \text{cm}^3$.
Why: A cylinder is a prism, so volume is cross-sectional area multiplied by length.
Method
<i>To 3 significant figures, the volume is</i> 926cm^3 .
Why: Round the unrounded volume once at the end.
Final answer
<i>volume :</i> 926cm^3

FINAL ANSWER

Original question 18

4MA1-1H Q22 Q22

ORIGINAL QUESTION**5 marks**

- 22 A solid is made from a cone and a hemisphere.

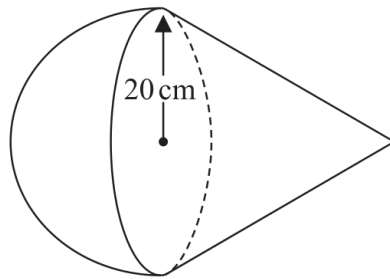


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{ cm}^2$

The volume of the solid is $k\pi\text{ cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

Worked solution 18

Chapter: Volume & Surface Area

MODULAR UNIT 2

5 marks

Q22 – Chapter: Volume & Surface Area

Family: Resolve composite solids and volume-unit constraints • Variant: Find volume of a joined composite solid

Method

For the cone, curved surface area $\pi r l = 580\pi$ gives $\pi(20)l = 580\pi$.

Why: Use the given radius in the curved-surface-area formula.

Method

The cone slant height is $l = 29$ cm.

Why: Cancel π and divide by 20.

Method

The cone height is $\sqrt{29^2 - 20^2} = \sqrt{441} = 21$ cm.

Why: The radius, height, and slant height form a right triangle.

Method

The solid volume is $(\frac{2}{3})\pi(20)^3 + (\frac{1}{3})\pi(20)^2(21)$.

Why: Add the hemisphere volume to the cone volume.

Method

This is $(\frac{16000}{3} + \frac{8400}{3})\pi = (\frac{24400}{3})\pi$, so $k = 24400/3$.

Why: Combine the exact coefficients of π .

Final answer

k: $k = 24400/3$

- 13 The diagram shows a solid cube.

The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

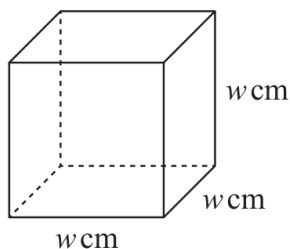


Diagram **NOT**
accurately drawn

The cube exerts a force of 56 newtons on the table.

The pressure on the table due to the cube is $0.14 \text{ newtons/cm}^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm^3

Worked solution

January 2020 | Question 13 | 4 marks

Family: Find volume of prisms and cylinders | Variant: Find volume of a cuboid or cylinder

METHOD / WORKING

$$1. \text{Area of contact} = 56/0.14 = 400\text{cm}^2.$$

$$2. \text{The square face gives } w^2 = 400, \text{ so } w = 20\text{cm}.$$

$$3. \text{Volume} = w^3.$$

$$4. V = 20^3 = 8000\text{cm}^3.$$

FINAL ANSWER

$$8000\text{cm}^3$$

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

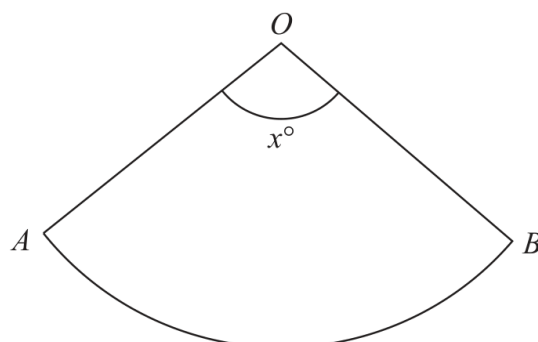


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

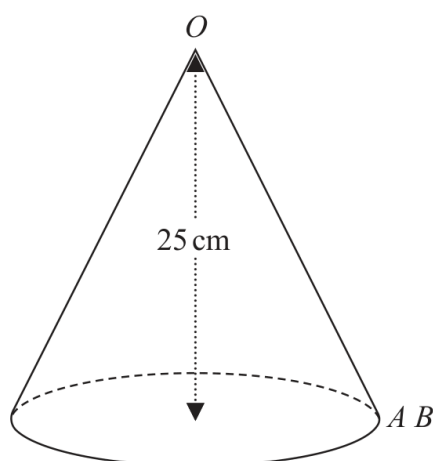


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm .

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

Worked solution

January 2020 | Question 26 | 6 marks

Family: Use sphere and cone formulae | Variant: Find cone surface area or recover its sector

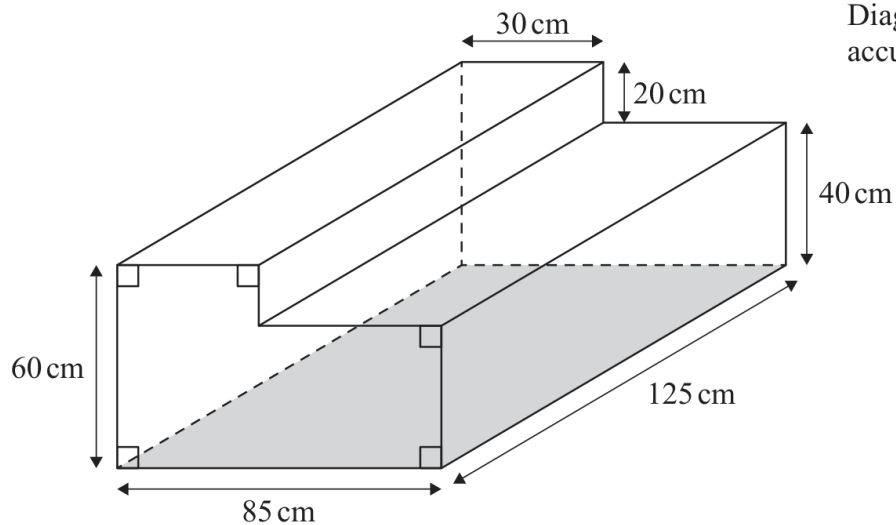
METHOD / WORKING

1. Cone volume gives $\pi r^2 l = 4800$, so $r = 7.8176\text{cm}$.
2. Slant height $l = \sqrt{r^2 + 25^2} = 26.1938\text{cm}$.
3. Sector arc equals base circumference: $(x/360)2\pi l = 2\pi r$.
4. Cancel 2π and rearrange $x = 360r/l$.
5. $x \sim 107.443^\circ$.
6. Nearest whole degree: $x = 107^\circ$.

FINAL ANSWER

107°

- 3 The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

Worked solution

January 2021 | Question 3 | 4 marks

Family: Find volume of prisms and cylinders | Variant: Resolve transfer, capacity or rate involving prism volume

METHOD / WORKING

1. $\text{Volumepumped in the first 90 minutes} = 30 \times 20 \times 125 = 75000 \text{ cm}^3$.

2. $\text{The full prism volume is } 500000 \text{ cm}^3$, so $\text{the full time is } 500000 / 75000 \times 90 = 600 \text{ minutes}$.

3. 600 minutes is 10 hours after 10:30.

4. All water is pumped out at 20:30.

FINAL ANSWER

20:30

17 The diagram shows a solid prism $ABCDEFGH$.

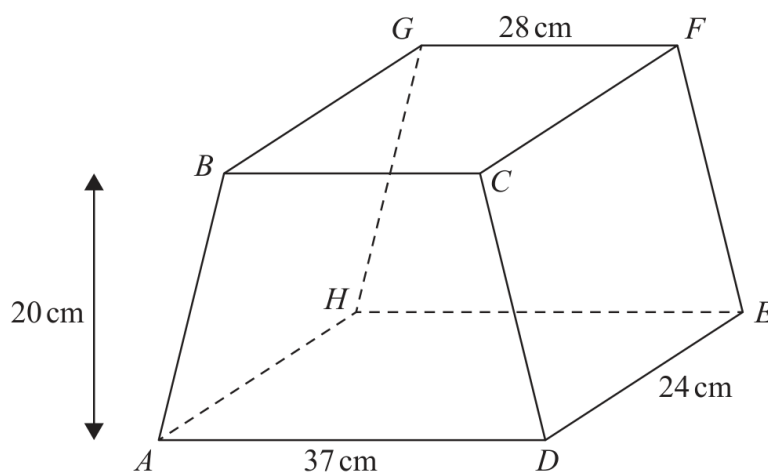


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm^2
(4)

Worked solution

January 2021 | Question 17 | 4 marks

Family: Find surface area of prisms and cylinders | Variant: Find surface area of a cuboid or simple prism

METHOD / WORKING

1. $\text{Area of trapezium } ABCD = \frac{1}{2}(37 + 28) \times 20 = 650 \text{ cm}^2.$
2. Sloping side $AB = \sqrt{20^2 + 4.5^2} = 20.5 \text{ cm}.$
3. Add two trapezia and four rectangular faces: $2 \times 650 + 2 \times 20.5 \times 24 + 37 \times 24 + 28 \times 24.$
4. $\text{Total surface area} = 3844 \text{ cm}^2.$

FINAL ANSWER 3844 cm^2

10 Here is a triangular prism.

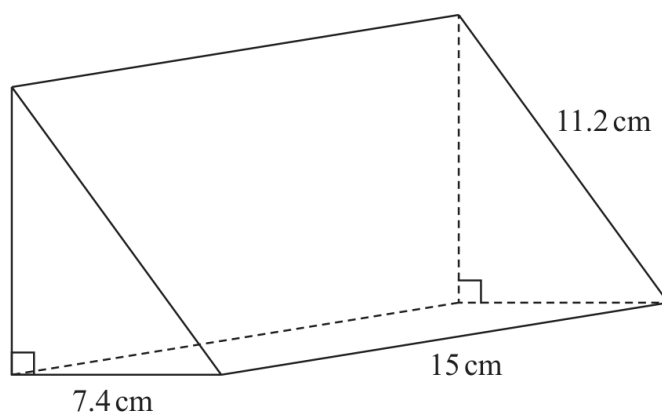


Diagram **NOT**
accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm³

Worked solution

June 2021 | Question 10 | 5 marks

Family: Find volume of prisms and cylinders | Variant: Find volume of a prism from its cross-section

METHOD / WORKING

1. Find the perpendicular height of the triangular cross-section: $\sqrt{11.2^2 - 7.4^2} = 8.407... \text{ cm}$.
2. *Cross – section area* $= \frac{1}{2} \times 7.4 \times 8.407... = 31.10... \text{ cm}^2$.
3. *Multiply by prism length* $15 : 31.10... \times 15 = 466.59... \text{ cm}^3$.
4. Round to 3 significant figures.
5. *State the volume as* 467 cm^3 .

FINAL ANSWER 467 cm^3

- 10 A solid aluminium cylinder has radius 10 cm and height h cm.

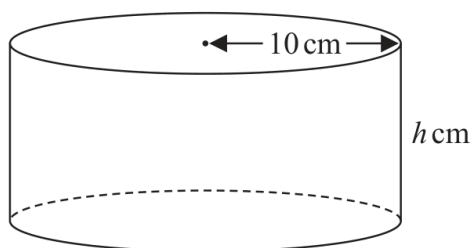


Diagram **NOT**
accurately drawn

The mass of the cylinder is 5.4 kg.

The density of aluminium is 0.0027 kg/cm^3

Calculate the value of h .

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

Worked solution

November 2021 | Question 10 | 5 marks

Family: Find volume of prisms and cylinders | Variant: Recover a prism or cylinder dimension from volume

METHOD / WORKING

1. The mass is 5.4kg , and the density is 0.0027kg/cm^3 .2. $\text{volume} = \frac{\text{mass}}{\text{density}}$ 3. $= \frac{5.4}{0.0027} = 2000\text{ cm}^3$

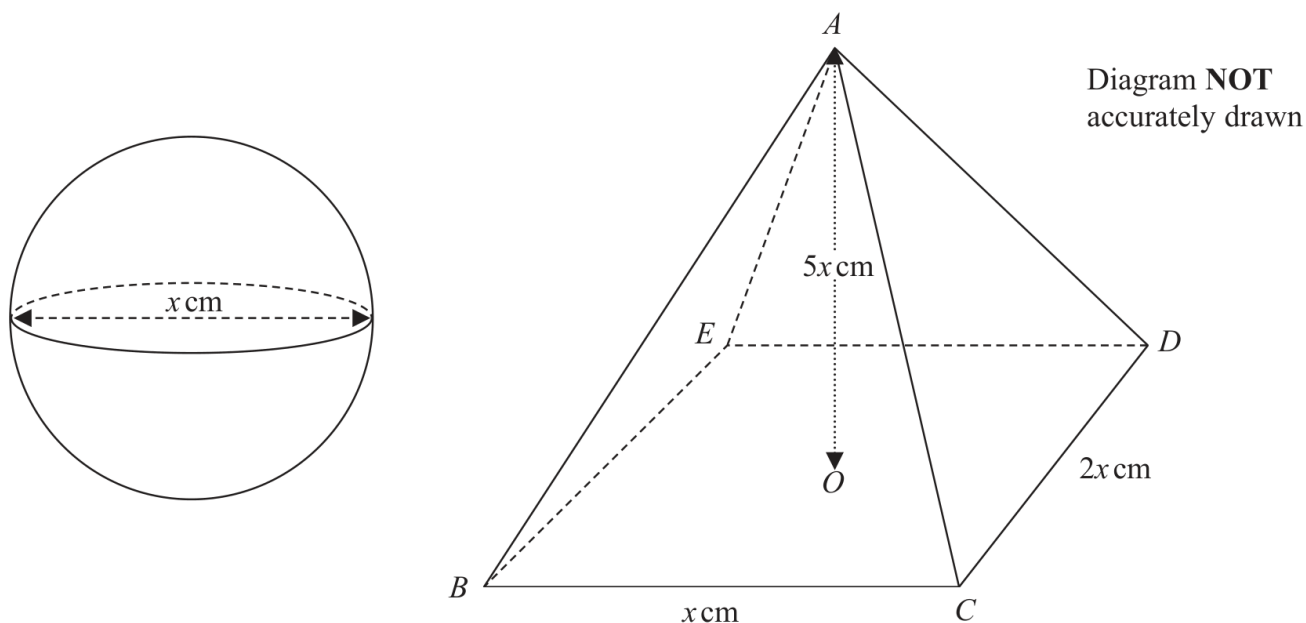
4. For the cylinder,

5. $V = \pi r^2 h$ 6. $2000 = \pi(10)^2 h$ 7. $h = \frac{2000}{100\pi} = 6.366\dots$ 8. Answer: 6.4cm .

FINAL ANSWER

6.4 cm.

- 22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is 288π cm³

Calculate the total surface area of the pyramid.

Give your answer correct to the nearest cm²

..... cm²

Worked solution

November 2021 | Question 22 | 6 marks

Family: Resolve composite solids and volume-unit constraints | Variant: Find volume of a joined composite solid

METHOD / WORKING

1. The sphere has diameter x , so its radius is
2. $\frac{x}{2}$
3. The volume of the sphere is $\frac{4}{3}\pi r^3$:
4. $\frac{4}{3}\pi\left(\frac{x}{2}\right)^3 = 288\pi$
5. $\frac{\pi x^3}{6} = 288\pi$
6. $x^3 = 1728$
7. $x = 12$
8. The rectangular base of the pyramid has dimensions x and $2x$, so its area is
9. $x(2x) = 2x^2$
10. For the two faces with base x , the slant height is
11. $\sqrt{(5x)^2 + x^2} = x\sqrt{26}$
12. Their total area is
13. $2\left(\frac{1}{2}x \cdot x\sqrt{26}\right) = x^2\sqrt{26}$
14. For the two faces with base $2x$, the slant height is
15. $\sqrt{(5x)^2 + (2x)^2}$
16. $= \frac{\sqrt{29}x}{2}$
17. Their total area is
18. $2\left(\frac{1}{2}(2x) \cdot \frac{\sqrt{29}x}{2}\right) = x^2\sqrt{29}$
19. Total surface area:
20. $2x^2 + x^2\sqrt{26} + x^2\sqrt{29}$
21. Substitute $x = 12$:
22. $144(2 + \sqrt{26} + \sqrt{29}) = 2469.440\dots$
23. Correct to the nearest cm^2 :

FINAL ANSWER

Worked solution

METHOD / WORKING

Worked solution (continued)
24. 2469
25. Answer: $\boxed{2469 \text{ cm}^2}$

FINAL ANSWER

 $\boxed{2469 \text{ cm}^2}$

- 9 The diagram shows a solid cylinder with radius 3 m.

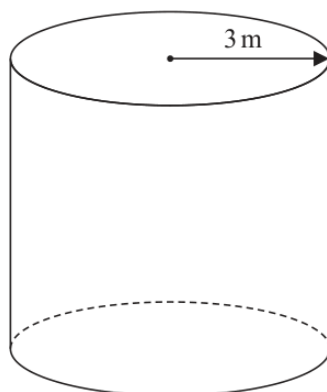


Diagram **NOT**
accurately drawn

The volume of the cylinder is $72\pi \text{ m}^3$

Calculate the **total** surface area of the cylinder.
Give your answer correct to 3 significant figures.

..... m^2

(Total for Question 9 is 5 marks)

Answer reference | May-Nov 2020 | Question 9

Family: Find surface area of prisms and cylinders • Variant: Recover a dimension from surface area

9	$\pi \times 3^2 \times h = 72\pi$ oe			M1	Allow use of 3.14... or $\frac{22}{7}$ for π and use of 226... for 72π
	$h = 72\pi \div (\pi \times 3^2)$ oe or $h = 8$			M1	method to isolate h (may be seen in several stages)
	$2 \times \pi \times 3^2 (= 18\pi \text{ or } 56.54\dots)$ or $2 \times \pi \times 3 \times "8"$ oe ($= 48\pi$ or 150 - 151)			M1	method to find the area of the two circles or curved surface area – use of their h , dep on 1st M1 (NB may get this mark for total area of 2 circles with no previous marks awarded)
	$2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times "8"$ oe ($= 66\pi$)			M1	method to find total surface area ft their h dep on 1st M1, including intention to add, to find the total surface area
		207	5	A1	accept 207-208
Total 5 marks					

Worked solution

May-Nov 2020 | Question 9 | 5 marks

Family: Find surface area of prisms and cylinders • Variant: Recover a dimension from surface area

Official final mark-scheme pages are embedded immediately after this question.

21 Given that the surface area of a sphere is $49\pi\text{cm}^2$

find the volume of the sphere.

Give your answer correct to the nearest integer.

..... cm^3

(Total for Question 21 is 3 marks)

Answer reference | January 2023 | Question 21

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Question	Working	Answer	Mark	Notes
21	$r = \sqrt{\frac{49\pi}{4\pi}}$ oe (= 3.5)		3	M1
	[volume =] $\frac{4}{3} \times \pi \times 3.5^3$			M1
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	180		A1 awrt 180
				Total 3 marks

Worked solution

January 2023 | Question 21 | 3 marks

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Official final mark-scheme pages are embedded immediately after this question.